4-IN-1 ESC / REFERENCE CIRCUITS

5V buck candidate

File: 00-buck.kicad_sch

Input, logic power and integration boundary

File: 01-power-interface.kicad_sch

MCU pair A

File: 02-controller-A.kicad_sch

Driver M1

File: 04-driver-M1.kicad_sch

Bridge M1

File: 08-bridge-M1.kicad_sch

Driver M2

File: 05-driver-M2.kicad_sch

Bridge M2

File: 09-bridge-M2.kicad_sch

MCU pair B

File: 03-controller-B.kicad_sch

Driver M3

File: 06-driver-M3.kicad_sch

Bridge M3

File: 10-bridge-M3.kicad_sch

Driver M4

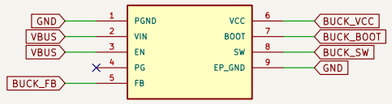
File: 07-driver-M4.kicad_sch

Bridge M4

File: 11-bridge-M4.kicad_sch

LMR36510ADDAR / 400kHz / NOMINAL 5V

U95
LMR36510ADDAR



L1
22u / Isat >=2A candidate



C1
100n / 16V



C2
1u / 10V



C3
2.2u effective / 100V



C4
220n / 100V



C5
22u / 25V



C6
22u / 25V



R1
80.4k / 0.1%



R2
20.0k / 0.1%



J1
VBAT / POWER GND
VBUS
GND
1
2

J2
5V TEST OUTPUT / NO BACKFEED
+5V
GND
1
2

C90
470u / 63V external candidate
VBUS
GND
+

D1
SMBJ30A / CANDIDATE ONLY
VBUS
GND

U90
AP2112K-3.3 / candidate
+5V
+5V
1
3
VIN
EN
VOUT
GND
5
2
3V3_A
GND

U91
AP2112K-3.3 / candidate
+5V
+5V
1
3
VIN
EN
VOUT
GND
5
2
3V3_B
GND

C91
1u
+5V
GND

C92
4.7u
3V3_A
GND

C93
1u
+5V
GND

C94
4.7u
3V3_B
GND

GND

VBUS

+5V

PAIR A: STM32G474RET6 / TWO MOTORS / HRTIM A..F

U1
STM32G474RET6

J10
DEBUG / PAIR A

J20
FC / PAIR A

C95
100n

C96
100n

C97
100n

C98
100n

C99
4.7u

C100
100n / VDDA

C101
1u / VREF

R3
10k

C102
100n

R4
10k

R5
150k / 0.1%

R6
10k / 0.1%

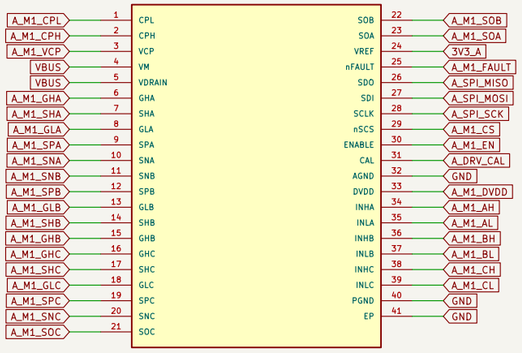
C103
10n

TP1
TIMING

R7
2.2k

D10
STATUS

MOTOR 1 / DRV8323S / SIX PWM / THREE LOW-SIDE SHUNTS























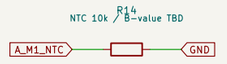


















MOTOR 1 / POWER CHANNEL / 60V MOSFET CANDIDATE

Q1
BSC028N06NS
A_M1_SHA
A_M1_SHA
A_M1_SHA
A_M1_A_GH
S
S
S
G
D
D
D
D
VBUS
VBUS
VBUS
VBUS

R18
4.7R / tune
A_M1_GHA
A_M1_A_GH

R19
100k
A_M1_A_GH
A_M1_SHA

Q3
BSC028N06NS
A_M1_SHB
A_M1_SHB
A_M1_SHB
A_M1_B_GH
VBUS
VBUS
VBUS
VBUS

R22
4.7R / tune
A_M1_GHB
A_M1_B_GH

R23
100k
A_M1_B_GH
A_M1_SHB

Q5
BSC028N06NS
A_M1_SHC
A_M1_SHC
A_M1_SHC
A_M1_C_GH
VBUS
VBUS
VBUS
VBUS

R26
4.7R / tune
A_M1_GHC
A_M1_C_GH

R27
100k
A_M1_C_GH
A_M1_SHC

Q2
BSC028N06NS
A_M1_LSA
A_M1_LSA
A_M1_LSA
A_M1_A_GL
A_M1_SHA
A_M1_SHA
A_M1_SHA
A_M1_SHA

R20
4.7R / tune
A_M1_GLA
A_M1_A_GL

R21
100k
A_M1_A_GL
A_M1_LSA

Q4
BSC028N06NS
A_M1_LSB
A_M1_LSB
A_M1_LSB
A_M1_B_GL
A_M1_SHB
A_M1_SHB
A_M1_SHB
A_M1_SHB

R24
4.7R / tune
A_M1_GLB
A_M1_B_GL

R25
100k
A_M1_B_GL
A_M1_LSB

Q6
BSC028N06NS
A_M1_LSC
A_M1_LSC
A_M1_LSC
A_M1_C_GL
A_M1_SHC
A_M1_SHC
A_M1_SHC
A_M1_SHC

R28
4.7R / tune
A_M1_GLC
A_M1_C_GL

R29
100k
A_M1_C_GL
A_M1_LSC

RS1
1m / WSKW06121L000FEA
A_M1_LSA
A_M1_SPA
A_M1_SNA
GND

RS2
1m / WSKW06121L000FEA
A_M1_LSB
A_M1_SPB
A_M1_SNB
GND

RS3
1m / WSKW06121L000FEA
A_M1_LSC
A_M1_SPC
A_M1_SNC
GND

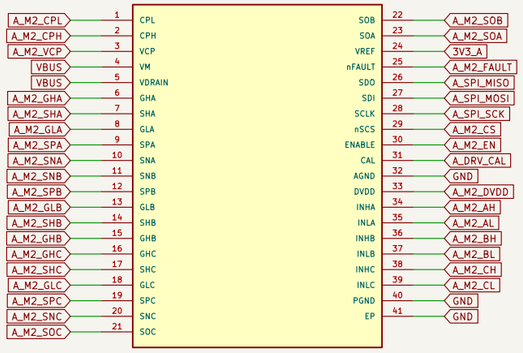
C113
4.7u / 100V / DC bias TBD
VBUS
GND

C114
4.7u / 100V / DC bias TBD
VBUS
GND

C115
4.7u / 100V / DC bias TBD
VBUS
GND

J31
M1 PHASES
A_M1_SHA
A_M1_SHB
A_M1_SHC

MOTOR 2 / DRV8323S / SIX PWM / THREE LOW-SIDE SHUNTS









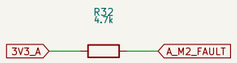
















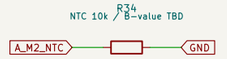












MOTOR 2 / POWER CHANNEL / 60V MOSFET CANDIDATE

Q7
BSC028N06NS
A_M2_SHA
A_M2_SHA
A_M2_SHA
A_M2_A_GH
VBUS
VBUS
VBUS
VBUS

R38
4.7R / tune
A_M2_GHA
A_M2_A_GH

R39
100k
A_M2_A_GH
A_M2_SHA

Q9
BSC028N06NS
A_M2_SHB
A_M2_SHB
A_M2_SHB
A_M2_B_GH
VBUS
VBUS
VBUS
VBUS

R42
4.7R / tune
A_M2_GHB
A_M2_B_GH

R43
100k
A_M2_B_GH
A_M2_SHB

Q11
BSC028N06NS
A_M2_SHC
A_M2_SHC
A_M2_SHC
A_M2_C_GH
VBUS
VBUS
VBUS
VBUS

R46
4.7R / tune
A_M2_GHC
A_M2_C_GH

R47
100k
A_M2_C_GH
A_M2_SHC

Q8
BSC028N06NS
A_M2_LSA
A_M2_LSA
A_M2_LSA
A_M2_A_GL
A_M2_SHA
A_M2_SHA
A_M2_SHA
A_M2_SHA

R40
4.7R / tune
A_M2_GLA
A_M2_A_GL

R41
100k
A_M2_A_GL
A_M2_LSA

Q10
BSC028N06NS
A_M2_LSB
A_M2_LSB
A_M2_LSB
A_M2_B_GL
A_M2_SHB
A_M2_SHB
A_M2_SHB
A_M2_SHB

R44
4.7R / tune
A_M2_GLB
A_M2_B_GL

R45
100k
A_M2_B_GL
A_M2_LSB

Q12
BSC028N06NS
A_M2_LSC
A_M2_LSC
A_M2_LSC
A_M2_C_GL
A_M2_SHC
A_M2_SHC
A_M2_SHC
A_M2_SHC

R48
4.7R / tune
A_M2_GLC
A_M2_C_GL

R49
100k
A_M2_C_GL
A_M2_LSC

A_M2_LSA
RS4
1m / WSKW06121L000FEA
A_M2_SPA
A_M2_SNA
GND

C125
4.7u / 100V / DC bias TBD
VBUS
GND

A_M2_LSB
RS5
1m / WSKW06121L000FEA
A_M2_SPB
A_M2_SNB
GND

C126
4.7u / 100V / DC bias TBD
VBUS
GND

A_M2_LSC
RS6
1m / WSKW06121L000FEA
A_M2_SPC
A_M2_SNC
GND

J32
M2 PHASES
A_M2_SHA
A_M2_SHB
A_M2_SHC

C127
4.7u / 100V / DC bias TBD
VBUS
GND

PAIR B: STM32G474RET6 / TWO MOTORS / HRTIM A..F

U2
STM32G474RET6

J11
DEBUG / PAIR B

J21
FC / PAIR B

C128
100n

C129
100n

C130
100n

C131
100n

C132
4.7u

C133
100n / VDDA

C134
1u / VREF

R50
10k

C135
100n

R51
10k

R52
150k / 0.1%

R53
10k / 0.1%

C136
10n

R54
2.2k

TP2
TIMING

D11
STATUS

MOTOR 3 / DRV8323S / SIX PWM / THREE LOW-SIDE SHUNTS

U13
DRV8323SRTAR

B_M1_CPL	1	CPL	
B_M1_CPH	2	CPH	
B_M1_VCP	3	VCP	
VBUS	4	VM	
VBUS	5	VDRAIN	
B_M1_GHA	6	GHA	
B_M1_SHA	7	SHA	
B_M1_GLA	8	GLA	
B_M1_SPA	9	SPA	
B_M1_SNA	10	SNA	
B_M1_SNB	11	SNB	
B_M1_SPB	12	SPB	
B_M1_GLB	13	GLB	
B_M1_SHB	14	SHB	
B_M1_GHB	15	GHB	
B_M1_GHC	16	GHC	
B_M1_SHC	17	SHC	
B_M1_GLC	18	GLC	
B_M1_SPC	19	SPC	
B_M1_SNC	20	SNC	
B_M1_SOC	21	SOC	
	22	SOB	B_M1_SOB
	23	SOA	B_M1_SOA
	24	VREF	3V3_B
	25	nFAULT	B_M1_FAULT
	26	SDO	B_SPI_MISO
	27	SDI	B_SPI_MOSI
	28	SCLK	B_SPI_SCK
	29	nSCS	B_M1_CS
	30	ENABLE	B_M1_EN
	31	CAL	B_DRV_CAL
	32	AGND	GND
	33	DVDD	B_M1_DVDD
	34	INHA	B_M1_AH
	35	INLA	B_M1_AL
	36	INHB	B_M1_BH
	37	INLB	B_M1_BL
	38	INHC	B_M1_CH
	39	INLC	B_M1_CL
	40	PGND	GND
	41	EP	GND

C137
47n / 63V
B_M1_CPH — B_M1_CPL

C138
1u / 25V
B_M1_VCP — VBUS

C139
1u / 10V
B_M1_DVDD — GND

C140
100n / 63V
VBUS — GND

C141
10u EFFECTIVE / 63V minimum
VBUS — GND

C142
100n / VREF
3V3_B — GND

R55
10k
3V3_B — B_M1_CS

R56
10k
B_M1_EN — GND

R57
4.7k
3V3_B — B_M1_FAULT

R58
4.7k / shared bus
3V3_B — B_SPI_MISO

R59
10k
B_DRV_CAL — GND

R62
47R / starting value
B_M1_SOA — B_M1_IA

R63
47R / starting value
B_M1_SOB — B_M1_IB

R64
47R / starting value
B_M1_SOC — B_M1_IC

R60
10k / NTC pullup
3V3_B — B_M1_NTC

R61
NTC 10k / B-value TBD
B_M1_NTC — GND

C143
470p / starting value
B_M1_IA — GND

C144
470p / starting value
B_M1_IB — GND

C145
470p / starting value
B_M1_IC — GND

MOTOR 3 / POWER CHANNEL / 60V MOSFET CANDIDATE

Q13
BSC028N06NS
B_M1_SHA
B_M1_SHA
B_M1_SHA
B_M1_A_GH
S
S
S
G
D
D
D
D
VBUS
VBUS
VBUS
VBUS

R65
4.7R / tune
B_M1_GHA
B_M1_A_GH

R66
100k
B_M1_A_GH
B_M1_SHA

Q15
BSC028N06NS
B_M1_SHB
B_M1_SHB
B_M1_SHB
B_M1_B_GH
VBUS
VBUS
VBUS
VBUS

R69
4.7R / tune
B_M1_GHB
B_M1_B_GH

R70
100k
B_M1_B_GH
B_M1_SHB

Q17
BSC028N06NS
B_M1_SHC
B_M1_SHC
B_M1_SHC
B_M1_C_GH
VBUS
VBUS
VBUS
VBUS

R73
4.7R / tune
B_M1_GHC
B_M1_C_GH

R74
100k
B_M1_C_GH
B_M1_SHC

Q14
BSC028N06NS
B_M1_LSA
B_M1_LSA
B_M1_LSA
B_M1_A_GL
B_M1_SHA
B_M1_SHA
B_M1_SHA
B_M1_SHA

R67
4.7R / tune
B_M1_GLA
B_M1_A_GL

R68
100k
B_M1_A_GL
B_M1_LSA

Q16
BSC028N06NS
B_M1_LSB
B_M1_LSB
B_M1_LSB
B_M1_B_GL
B_M1_SHB
B_M1_SHB
B_M1_SHB
B_M1_SHB

R71
4.7R / tune
B_M1_GLB
B_M1_B_GL

R72
100k
B_M1_B_GL
B_M1_LSB

Q18
BSC028N06NS
B_M1_LSC
B_M1_LSC
B_M1_LSC
B_M1_C_GL
B_M1_SHC
B_M1_SHC
B_M1_SHC
B_M1_SHC

R75
4.7R / tune
B_M1_GLC
B_M1_C_GL

R76
100k
B_M1_C_GL
B_M1_LSC

RS7
1m / WSKW06121L000FEA
B_M1_LSA
B_M1_SPA
B_M1_SNA
GND

C146
4.7u / 100V / DC bias TBD
VBUS
GND

RS8
1m / WSKW06121L000FEA
B_M1_LSB
B_M1_SPB
B_M1_SNB
GND

C147
4.7u / 100V / DC bias TBD
VBUS
GND

RS9
1m / WSKW06121L000FEA
B_M1_LSC
B_M1_SPC
B_M1_SNC
GND

J33
M3 PHASES
B_M1_SHA
B_M1_SHB
B_M1_SHC

C148
4.7u / 100V / DC bias TBD
VBUS
GND

MOTOR 4 / DRV8323S / SIX PWM / THREE LOW-SIDE SHUNTS

U14
DRV8323SRTAR

B_M2_CPL	1	CPL	SOB	22	B_M2_SOB
B_M2_CPH	2	CPH	SOA	23	B_M2_SOA
B_M2_VCP	3	VCP	VREF	24	3V3_B
VBUS	4	VM	nFAULT	25	B_M2_FAULT
VBUS	5	VDRAIN	SDO	26	B_SPI_MISO
B_M2_GHA	6	GHA	SDI	27	B_SPI_MOSI
B_M2_SHA	7	SHA	SCLK	28	B_SPI_SCK
B_M2_GLA	8	GLA	nSCS	29	B_M2_CS
B_M2_SPA	9	SPA	ENABLE	30	B_M2_EN
B_M2_SNA	10	SNA	CAL	31	B_DRV_CAL
B_M2_SNB	11	SNB	AGND	32	GND
B_M2_SPB	12	SPB	DVDD	33	B_M2_DVDD
B_M2_GLB	13	GLB	INHA	34	B_M2_AH
B_M2_SHB	14	SHB	INLA	35	B_M2_AL
B_M2_GHB	15	GHB	INHB	36	B_M2_BH
B_M2_GHC	16	GHC	INLB	37	B_M2_BL
B_M2_SHC	17	SHC	INHC	38	B_M2_CH
B_M2_GLC	18	GLC	INLC	39	B_M2_CL
B_M2_SPC	19	SPC	PGND	40	GND
B_M2_SNC	20	SNC	EP	41	GND
B_M2_SOC	21	SOC			

C149
47n / 63V
B_M2_CPH — B_M2_CPL

C150
1u / 25V
B_M2_VCP — VBUS

C151
1u / 10V
B_M2_DVDD — GND

C152
100n / 63V
VBUS — GND

C153
10u EFFECTIVE / 63V minimum
VBUS — GND

C154
100n / VREF
3V3_B — GND

R77
10k
3V3_B — B_M2_CS

R78
10k
B_M2_EN — GND

R79
4.7k
3V3_B — B_M2_FAULT

R82
47R / starting value
B_M2_SOA — B_M2_IA

R83
47R / starting value
B_M2_SOB — B_M2_IB

R84
47R / starting value
B_M2_SOC — B_M2_IC

R80
10k / NTC pullup
3V3_B — B_M2_NTC

R81
NTC 10k / B-value TBD
B_M2_NTC — GND

C155
470p / starting value
B_M2_IA — GND

C156
470p / starting value
B_M2_IB — GND

C157
470p / starting value
B_M2_IC — GND

MOTOR 4 / POWER CHANNEL / 60V MOSFET CANDIDATE

Q19
BSC028N06NS
B_M2_SHA
B_M2_SHA
B_M2_SHA
B_M2_A_GH
S S S G D D D D
VBUS VBUS VBUS VBUS

R85
4.7R / tune
B_M2_GHA
B_M2_A_GH

R86
100k
B_M2_A_GH
B_M2_SHA

Q21
BSC028N06NS
B_M2_SHB
B_M2_SHB
B_M2_SHB
B_M2_B_GH
S S S G D D D D
VBUS VBUS VBUS VBUS

R89
4.7R / tune
B_M2_GHB
B_M2_B_GH

R90
100k
B_M2_B_GH
B_M2_SHB

Q23
BSC028N06NS
B_M2_SHC
B_M2_SHC
B_M2_SHC
B_M2_C_GH
S S S G D D D D
VBUS VBUS VBUS VBUS

R93
4.7R / tune
B_M2_GHC
B_M2_C_GH

R94
100k
B_M2_C_GH
B_M2_SHC

Q20
BSC028N06NS
B_M2_LSA
B_M2_LSA
B_M2_LSA
B_M2_A_GL
S S S G D D D D
B_M2_SHA B_M2_SHA B_M2_SHA B_M2_SHA

R87
4.7R / tune
B_M2_GLA
B_M2_A_GL

R88
100k
B_M2_A_GL
B_M2_LSA

Q22
BSC028N06NS
B_M2_LSB
B_M2_LSB
B_M2_LSB
B_M2_B_GL
S S S G D D D D
B_M2_SHB B_M2_SHB B_M2_SHB B_M2_SHB

R91
4.7R / tune
B_M2_GLB
B_M2_B_GL

R92
100k
B_M2_B_GL
B_M2_LSB

Q24
BSC028N06NS
B_M2_LSC
B_M2_LSC
B_M2_LSC
B_M2_C_GL
S S S G D D D D
B_M2_SHC B_M2_SHC B_M2_SHC B_M2_SHC

R95
4.7R / tune
B_M2_GLC
B_M2_C_GL

R96
100k
B_M2_C_GL
B_M2_LSC

B_M2_LSA
RS10
1m / WSKW06121L000FEA
1 2 3 4
B_M2_SPA
B_M2_SNA
GND

B_M2_LSB
RS11
1m / WSKW06121L000FEA
1 2 3 4
B_M2_SPB
B_M2_SNB
GND

B_M2_LSC
RS12
1m / WSKW06121L000FEA
1 2 3 4
B_M2_SPC
B_M2_SNC
GND

J34
M4 PHASES
B_M2_SHA
B_M2_SHB
B_M2_SHC
1 2 3

C158
4.7u / 100V / DC bias TBD
VBUS GND

C159
4.7u / 100V / DC bias TBD
VBUS GND

C160
4.7u / 100V / DC bias TBD
VBUS GND